

28. Ans: A

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\frac{V_s}{120} = \frac{1}{60}$$

$$V_s = 2.0 \text{ V}$$

$$\text{Then } I_s = V_s/R = 2.0/3.0 = 0.667 \text{ } \Omega$$

For ideal transformer,

$$V_p I_p = V_s I_s$$

$$120 \times I_p = 2.0 \times 0.667$$

$$I_p = 0.011 \text{ A}$$